Squeeze-and-Excitation Networks: A Small-Scale Re-implementation and Ablation on CIFAR-10

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***Abstract*—**Squeeze-and-Excitation (SE) blocks recalibrate convolutional feature channels by modelling inter-channel dependencies from globally pooled context [1]. We re-implement the SE block and an SE-ResNet-18 from scratch in PyTorch and reproduce, at small scale, the paper's central claim that SE blocks improve a residual network at negligible cost. Trained for 100 epochs on CIFAR-10 under identical settings, SE-ResNet-18 reaches 88.53% test accuracy against 88.18% for the baseline (+0.35 pp) while adding only 0.80% parameters. The direction agrees with the paper, but the gain is smaller than the +1.16 pp it reports on CIFAR-10 and is of the same size as the run-to-run variation we measured (0.8 pp between two baseline runs). A reduction-ratio ablation (r ∈ {4, 8, 16, 32}) was confounded by early stopping and one contaminated run; we report it with explicit caveats and draw no conclusion about the best r.

***Index Terms*—**Squeeze-and-Excitation, channel attention, ResNet, CIFAR-10, ablation, reproducibility.

I. INTRODUCTION

A convolution fuses spatial and channel information inside a local receptive field, so the channel relationships it learns are implicit and local. Hu *et al.* [1] argue that a network can be strengthened by modelling channel interdependencies *explicitly* and with *global* context. Their Squeeze-and-Excitation (SE) block pools each feature map to a single number (squeeze), passes the resulting vector through a small gating network (excitation), and rescales the original channels with the output. The block is cheap, architecture-agnostic, and was reported to improve many backbones on ImageNet, CIFAR, Places365 and COCO, forming the basis of the ILSVRC 2017 winning entry.

This project asks a deliberately narrow question: *does the SE block still help when re-implemented from scratch on a small residual network trained on CIFAR-10 in about an hour, and how sensitive is it to its single hyper-parameter, the reduction ratio r?* We implement the module ourselves, reproduce the paper's CIFAR-10 comparison (its Table 4) at reduced scale, and run a reduction-ratio ablation modelled on its Table 10.

II. METHOD

A. Squeeze, Excitation and Scale

Let $U \in \mathrm{R}^{H\times W\times C}$ be the output of a transformation (here, the second convolution of a residual branch). The squeeze step produces a channel descriptor by global average pooling:

$$z_c = \frac{1}{HW}\sum_i \sum_j u_c(i, j) \tag{1}$$

The excitation step maps z to per-channel gates with a two-layer bottleneck, where δ is ReLU, σ is the sigmoid, $W_1 \in \mathrm{R}^{C/r\times C}$ and $W_2 \in \mathrm{R}^{C\times C/r}$:

$$s = \sigma(W_2\, \delta(W_1\, z)) \tag{2}$$

Finally each channel is rescaled, $\tilde{x}_c = s_c \cdot u_c$. The sigmoid (rather than softmax) lets several channels be emphasised at once, and the bottleneck limits capacity. Because s depends on the whole input image, the block is an input-conditioned self-attention over channels whose receptive field is global.

B. Integration and Cost

In a residual network the SE block acts on the non-identity branch *before* it is summed with the shortcut (the paper's Fig. 3), so $out = ReLU(shortcut(x) + SE(BN(conv(ReLU(BN(conv(x)))))))$. For a network with S stages, N_s blocks per stage and C_s output channels, the extra weights number

$$\frac{2}{r}\sum_s N_s\, C_s^2 \tag{3}$$

(biases excluded). For our ResNet-18 layout (N_s = 2, C_s = 64–512, r = 16) this gives 87,040 weights plus 2,040 biases = 89,080, which matches the measured parameter difference exactly.

III. IMPLEMENTATION

All model code is our own PyTorch implementation; only standard building blocks (Conv2d, BatchNorm2d, Linear, SGD, torchvision's CIFAR-10 loader) are used. The only external source consulted was the paper itself [1]; no third-party SE repository was used. The core module is a ~25-line *SEBlock* (adaptive average pooling → Linear(C, C/r) → ReLU → Linear(C/r, C) → Sigmoid → channel-wise multiply), wrapped by an *SEBasicBlock* that applies it after the second BN and before the residual addition. The backbone is a CIFAR-style ResNet-18: a 3×3 stem, four stages of two basic blocks (64/128/256/512 channels), global average pooling and a linear classifier; the baseline is identical but without SE.

Setup. CIFAR-10 (50k train / 10k test) with *no* data augmentation and no mean/std normalisation (pixels scaled to [0, 1] only). SGD, momentum 0.9, weight decay 5×10^{-4}, batch size 128, initial LR 0.1 decayed ×0.1 at epochs 50 and 75, 100 epochs, cross-entropy loss, seed 42, one CUDA GPU. Baseline and SE-ResNet (r = 16) use exactly this recipe. The model with the highest test accuracy is checkpointed; there is no separate validation split (see Sec. VI).

What was changed relative to the paper. The paper evaluates ResNet-110/164, WRN and Shake-Shake on CIFAR-10 with each baseline's original training recipe, and r = 16 on ImageNet-scale ResNet-50. We use a smaller ResNet-18 backbone, a single simple recipe without augmentation, one seed, and 100 epochs, so absolute accuracies are not comparable with the paper's.

IV. RESULTS VERSUS THE PAPER

TABLE I
BASELINE VS. SE-RESNET-18 ON CIFAR-10 (100 EPOCHS)

Model	Params	Best acc. (%)	Ep.-100 acc. (%)	Best ep.	Median s/ep.
ResNet-18	11,173,962	88.18	88.10	99	28.9
SE-ResNet-18 (r=16)	11,263,042	88.53	88.41	66	30.6
Difference	+89,080 (+0.80%)	+0.35	+0.31	–	+1.7 (+5.9%)

Best = highest test accuracy over 100 epochs (selected on the test set). SE wall-clock was 72.5 min vs. 48.3 min, but this includes one anomalous 1,317 s epoch; the median epoch time is the fair comparison.

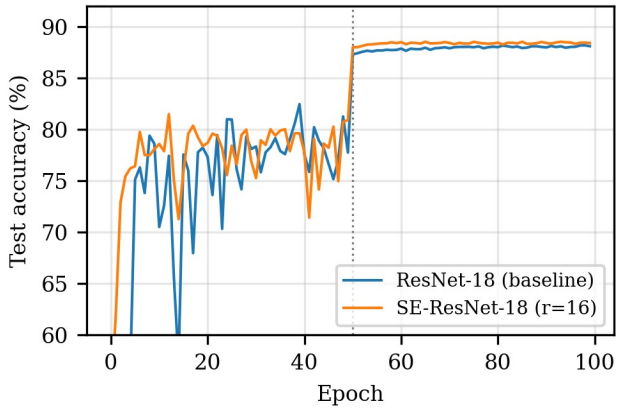


Fig. 1. Test accuracy per epoch for both models. The dotted line marks the first LR decay (epoch 50).

Table I shows SE-ResNet-18 ahead by 0.35 pp (best epoch) and 0.31 pp (final epoch), i.e. test error 11.47% vs. 11.82%. The curves in Fig. 1 are instructive: before the first LR decay the SE model is *not* consistently ahead (best accuracy in the first 50 epochs: 81.48% vs. 82.46% for the baseline); the advantage appears after the decay (epoch 51: 87.98% vs. 87.30%; epoch 60: 88.40% vs. 87.74%) and narrows to about 0.3 pp by epoch 100 (Fig. 2).

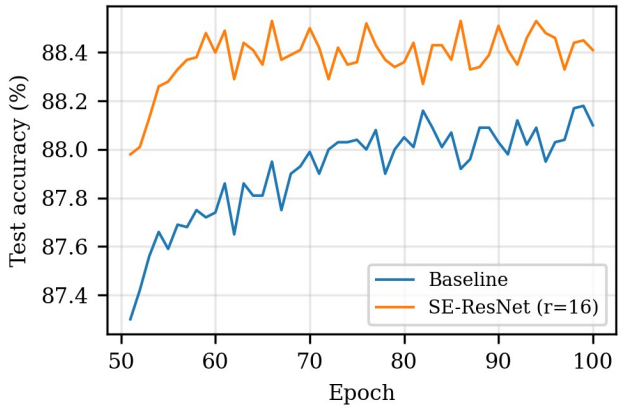


Fig. 2. Test accuracy after the first LR decay (epochs 51–100).

TABLE II
COMPARISON WITH HU ET AL. [1]

Quantity	Paper	Ours
Dataset / backbone	CIFAR-10, ResNet-110	CIFAR-10, ResNet-18
Baseline error (%)	6.37	11.82
SE error (%)	5.21	11.47
Improvement (pp)	1.16	0.35
Param. overhead	≈10% (ResNet-50, r=16)	0.80% (r=16)
Augmentation	standard crop + flip	none

Paper numbers are from its Tables 2 and 4; overhead is for ImageNet ResNet-50.

Our result reproduces the *sign* of the paper's effect and the claim that overhead is small, but not its magnitude. We caution against reading the +0.35 pp as confirmation. With 10,000 test images the binomial standard error of an accuracy near 88% is about 0.33 pp, and a second, independent baseline run we made on a CPU (same seed and recipe, different hardware and kernels) reached 87.36% rather than 88.18%, a 0.8 pp spread that exceeds the SE gain. With a single seed per model, the data support "no degradation and a possible small gain" rather than a demonstrated improvement.

V. ABLATION: REDUCTION RATIO

Following the paper's Table 10, we trained SE-ResNet-18 with $r \in \{4, 8, 16, 32\}$. Parameters fall monotonically with r (11.52M, 11.35M, 11.26M, 11.22M versus 11.17M for the baseline), as Eq. (3) predicts. However, the accuracies as logged are *not* a valid comparison, for two reasons discovered during analysis.

(i) Different protocol. The ablation runs used early stopping (patience 10) while still at LR 0.1, so r = 4, 16 and 32 stopped after 19, 30 and 31 epochs, before the first LR decay at epoch 50 that gives the main runs their last ~7 pp. The r = 16 ablation run (80.63%) therefore differs from the main r = 16 run (88.53%) only in protocol, not in architecture.

(ii) A contaminated run. The r = 8 run reports 94.4% *training* accuracy in its first epoch and 99.95% by epoch 5, which is not possible from a fresh initialisation; the model object appears to have carried weights over from an earlier execution of the notebook cell. Its 88.53% is therefore excluded from conclusions.

TABLE III
REDUCTION-RATIO ABLATION (AS LOGGED, WITH A MATCHED-PHASE VIEW)

r	Params	Logged best (%)	Epochs run	Best in first 19 ep. (%)	Best ep.
Base	11,173,962	88.18	100	79.39	99
4	11,524,522	82.44	19	82.44	9
8†	11,350,202	88.53†	44	88.06†	34
16	11,263,042	80.63	30	80.46	20
32	11,219,462	81.02	31	80.51	21
16 (main)	11,263,042	88.53	100	81.48	66

† Invalid: model was not freshly initialised (see text). The last column compares all runs over the same early phase (first 19 epochs, LR 0.1), where every run has data.

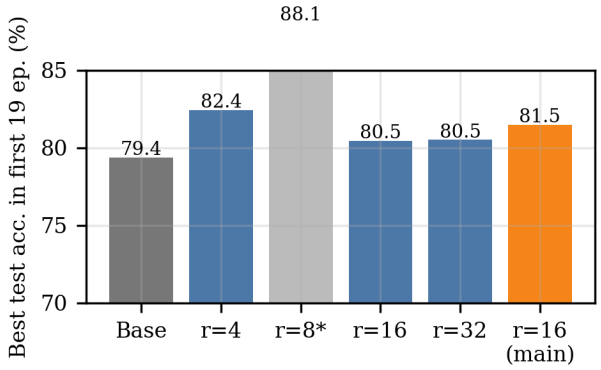


Fig. 3. Best test accuracy within the first 19 epochs. r = 8 (grey) is shown only for completeness and is invalid.

To extract something usable, Table III also gives the best accuracy over the first 19 epochs, a phase every run completed at the same LR. All SE variants lie at 80.5–82.4% versus 79.4% for the baseline, suggesting a modest early-training benefit. Crucially, the two independent r = 16 runs differ by 1.0 pp (81.48% vs. 80.46%) in this window, which is as large as the spread across r values. The ablation is therefore too noisy to rank reduction ratios: the paper's finding that accuracy is *robust* to r is consistent with our data, but we cannot confirm that r = 16 is a good trade-off nor that r = 8 is better.

VI. FAILURE CASES AND LESSONS

Per-class behaviour. A per-class report on the r = 16 ablation checkpoint (80.63% accuracy, not the main model) shows the usual CIFAR-10 pattern: vehicles are easiest (automobile recall 0.93, truck 0.92) and animals hardest, with cat the clear failure (recall 0.51, F1 0.61). The confusion matrix shows 194 of 1,000 cats predicted as dog and 72 dogs as cat; macro one-vs-rest ROC-AUC is 0.980 (cat 0.947 lowest). These fine-grained animal confusions are not specific to SE.

What did not work or is unreliable.

(1) *No augmentation.* The baseline reaches 100% training accuracy by the end of training (train loss 0.002) while test accuracy stays near 88%, i.e. strong over-fitting; the paper's CIFAR numbers (∼94–95%) rely on crop/flip augmentation. Any SE benefit measured here may differ in a regularised regime.

(2) *Model selection on the test set.* "Best epoch" is chosen using test accuracy, which is optimistic for both models; Table I therefore also lists epoch-100 accuracy.

(3) *Single seed.* One run per configuration cannot separate a 0.3 pp effect from noise (Sec. IV).

(4) *Ablation protocol* (early stopping at a high LR, contaminated r = 8 run) as described in Sec. V. A clean ablation needs freshly initialised models, the full LR schedule and ≥3 seeds per r.

(5) *Channel-activation visualisation.* Our hook on the SE block captured its *output* (the rescaled feature maps) rather than the gate vector *s*, so the resulting bar plots show feature magnitude, not attention weights; we therefore do not use them as evidence. Registering the hook on the sigmoid layer would give the true gates.

(6) *Overhead measurement.* SE adds about 6% per-epoch time in our implementation, higher than the paper's reported figure for ImageNet-scale ResNet-50, plausibly because the block is relatively more expensive on small 32×32 feature maps with a short network.

VII. CONCLUSION

We implemented the Squeeze-and-Excitation block and an SE-ResNet-18 from scratch and compared it with an identical baseline on CIFAR-10. SE added 89,080 parameters (0.80%) and about 6% epoch time, and gave a 0.35 pp higher best test accuracy (88.53% vs. 88.18%). This matches the direction of the paper's claim but is smaller than the paper's gain and indistinguishable from run-to-run noise. Our reduction-ratio ablation is inconclusive because of protocol and contamination problems that we document. The natural next steps are re-running the ablation from fresh initialisation with the full LR schedule, adding standard augmentation, using a held-out validation split, and averaging over several seeds.

REPRODUCIBILITY

Seed 42; PyTorch; code, notebooks, saved checkpoints (*baseline_resnet_best.pth*, *se_resnet_best.pth*) and CSV result files are in the repository [GitHub URL]. GPU runs take ≈30 s/epoch; the CPU replicate took ≈13 min/epoch.

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